Cross-species ncRNA annotation using synteny-constrained embedding similarity

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Abstract

Reference-based annotation transfer has been highly successful for protein-coding genes but remains difficult for non-coding RNAs (ncRNAs), which often show weak sequence conservation, ambiguous transcript boundaries, and incomplete cross-species annotation. Here we present CURIA, a pipeline for cross-species ncRNA annotation that combines genome alignment chains with representations from the RiNALMo RNA foundation model. CURIA constrains candidate loci using a chain-based projection inspired by TOGA and then identifies and matches localized embedding-supported regions within the projected loci. We applied CURIA to 19 human-to-query genome pairs. In a detailed human–mouse comparison, 52.4% of predicted short-ncRNA loci overlapped an annotated exon. For compact short ncRNAs, embedding similarity added little beyond conventional sequence metrics, whereas localized island matching provided complementary ranking information in human–mouse and human–cow comparisons. A multi-species screen further identified 1,693 human lncRNA loci containing recurrent embedding-supported regions in at least 17 of 19 query genomes, including a more stringent subset of 170 loci. These results establish a proof of concept for synteny-constrained, embedding-based ncRNA annotation and suggest that localized RNA embeddings can complement conventional sequence conservation when prioritizing candidate cross-species correspondences.

1 Introduction

1.1 Protein-coding orthology as a tractable problem

Orthology inference for protein-coding genes is well established and supported by scalable annotation transfer pipelines [1–7]. Strong sequence conservation, together with constraints imposed by open reading frames, enables reliable cross-species mapping of gene loci [2, 4]. Methods such as TOGA, which combine genome alignment chains with coding sequence structure, achieve high accuracy at genome scale. However, these approaches rely on properties specific to protein-coding genes and do not generalize to non-coding RNAs. In particular, they assume conserved coding structure and detectable sequence similarity, which are often absent outside protein-coding contexts.

In this work, we do not attempt to infer evolutionary orthology. Instead, we identify candidate cross-species correspondences supported by genomic synteny and embedding similarity, and frame the task as synteny-constrained annotation transfer.

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1.2 The ncRNA orthology challenge

Cross-species correspondence is difficult to establish for ncRNAs because the relevant conserved unit is often unclear. For many ncRNAs, evolutionary conservation is less readily captured by primary-sequence identity than for protein-coding genes: related RNAs may retain similar molecular functions, structural features, or localized conserved elements despite substantial sequence divergence [8–13]. Conservation may also be restricted to short motifs or local domains rather than extending across the complete transcript. At the same time, transcript boundaries and isoform structures are frequently ambiguous, and ncRNA annotations remain incomplete and heterogeneous across species [14, 15]. Together, these properties complicate both annotation transfer and its evaluation.

These problems are especially acute for lncRNAs. Annotated loci may span tens of kilobases and contain localized candidate domains separated by long intervening sequence. Whole-locus comparison can therefore dilute localized correspondence, and complete loci may exceed the input length supported by current RNA foundation models. This motivates searching for candidate conserved subregions rather than requiring end-to-end transcript correspondence.

1.3 Limitations of existing approaches

Existing methods address several related tasks, but none directly combines reference-guided annotation transfer with localization and comparison of candidate ncRNA subregions within syntenic genomic loci.

Covariance-model approaches implemented in Infernal and distributed through Rfam can sensitively identify members of established RNA families [16, 17]. Their application, however, depends on a predefined family alignment and consensus sequence–structure model. They are therefore well suited to annotating known RNA families, but less applicable to reference loci for which no curated family model exists.

Structure-aware alignment methods such as LocARNA and RNAforester compare supplied RNA sequences using explicit sequence–structure models [18, 19]. These methods address the comparison of already defined RNA intervals, rather than locating a corresponding element within a genome-scale syntenic search region.

Comparative screening methods such as RNAz and EvoFold identify genomic regions with evidence of evolutionarily conserved RNA secondary structure in multiple sequence alignments [20, 21]. Their objective is de novo discovery of candidate structured RNA elements, rather than transfer of a specific reference annotation to a query genome. They also depend on detectable comparative sequence and structural signals within the input alignment.

CURIA addresses a different task. Starting from a reference ncRNA annotation and pairwise genome alignment chains, it first restricts the search to candidate syntenic loci and then localizes and ranks corresponding subregions using learned RNA representations. This formulation does not require a pre-existing RNA family model, a supplied query transcript, or end-to-end conservation of the annotated reference locus.

1.4 Synteny-constrained embedding matching

An underexplored strategy is to use pretrained RNA-language-model representations as a localized similarity signal within syntenically constrained genomic search regions. Genome alignment chains provide candidate regions that preserve genomic context and therefore indicate where a corresponding ncRNA may be sought [22, 23]. RNA foundation models such as RNA-FM [24], RiNALMo [25], and related approaches such as RNABERT [26] provide learned representations associated with RNA family identity, secondary-structure-related properties, and downstream predictive tasks [27]. Applying these models as unconstrained genome-wide scanners would be computationally impractical, whereas restricting comparison to chain-defined candidate loci makes localized RNA-specific matching feasible. This combination supports comparison of ncRNA loci in settings where primary-sequence conservation is weak or heterogeneous.

Illustrative properties of the embedding representation and local-search behavior are shown in Figure 1.

1.5 Method overview

We formulate cross-species ncRNA correspondence as the matching of embedding-supported regions within loci defined by genome alignment chains. Candidate loci are first identified using a chain-based projection adapted from principles used in TOGA. The projection uses alignment coverage across exonic, intronic, and flanking sequence to provide locus-level syntenic context that is not determined solely by conservation within the ncRNA itself [1, 2].

Within each candidate locus, CURIA compares RNA-language-model representations using one of two geometry-dependent branches. Compact, single-exon ncRNAs are compared as complete regions using Maximum Mean Discrepancy (MMD) [28], which measures the difference between the distributions of their per-position embedding vectors. Longer or multi-exon loci are processed by a localized-subregion workflow, in which detector-positive subregions are matched by Smith–Waterman local alignment [29] over a per-token cosine-similarity matrix. Both branches provide continuous scores for ranking candidate correspondences without requiring nucleotide sequence alignment.

1.6 Contributions

In this work, we:

1. Formulate cross-species ncRNA annotation as synteny-constrained matching in RNA embedding space.
2. Introduce CURIA, a pipeline that combines chain-based locus projection with localized embedding-based comparison.
3. Develop complementary strategies for compact and extended ncRNA loci, including boundary refinement for short ncRNAs and localized subregion matching for longer or multi-exon loci.
4. Demonstrate the approach at genome scale across 19 mammalian genomes and identify recurrent candidate correspondences that are not fully explained by conventional sequence-similarity measures.

2 Methods

2.1 Pipeline overview

CURIA is a computational pipeline for genome-scale cross-species ncRNA correspondence that combines chain-guided locus prediction with embedding-based comparison (Figure 2). Given a reference genome with ncRNA annotations, a query genome, and pairwise genome alignment chains, CURIA produces refined query loci for compact single-exon ncRNAs and scored reference–query subregion correspondences for longer or multi-exon loci.

The pipeline consists of seven stages:

1. **Preprocessing.** Isoforms assigned to the same locus and strand are merged into a *union exon model* by taking the union of their exonic intervals in genomic order.

2. **Candidate region prediction.** Genome alignment chains are used to identify candidate query regions with a chain-based classifier adapted from principles used in TOGA (Section 2.3).

3. **Short ncRNA refinement.** For compact single-exon ncRNAs, the projected candidate locus is refined by an embedding-based local boundary search, without decomposition into subregions (Section 2.5).

4. **Reference subregion detection.** For longer or multi-exon loci, localized embedding-supported subregions ("islands") are identified in the reference locus (Section 2.7).

5. **Chain-guided subregion projection.** Each reference island is projected through the genome alignment chain to define a corresponding search region in the query genome (Section 2.8).

6. **Query subregion detection.** Candidate query islands are identified within the projected search regions associated with the reference islands (Section 2.9).

7. **Reference–query subregion matching.** Reference and query islands are compared by embedding-based local matching, producing scored reference–query subregion correspondences (Section 2.10).

Branch routing. Loci are assigned to processing branches according to the geometry of their union exon models rather than their annotation biotypes, because biotype labels do not reliably determine whether a locus is best treated as a compact complete region or as a set of localized subregions. Some annotated lncRNAs are short and single-exonic, whereas some compact ncRNA classes are multi-exonic.

Single-exon loci of length ≤ 256 nt are processed as complete loci. Single-exon loci of length 257–320 nt are represented as a single whole-exon island. Single-exon loci longer than 320 nt and all multi-exon loci are scanned for detector-positive islands (Section 2.7). These categories form a partition, and each locus is processed exactly once.

The union exon model is a locus-level search representation rather than a reconstructed mature transcript. Because exons are unioned across isoforms, the model may combine segments that do not coexist in any single transcript. It is used only to define the search space for localized signal.

2.2 Input data

CURIA requires three types of input data:

1. **Reference genome and annotation.** The reference genome is provided in 2bit format, with ncRNA annotations in BED12 format including `transcript_id`, `gene_id`, and `biotype`. Protein-coding transcripts are used during candidate-region prediction to compute synteny-related features but are excluded from downstream ncRNA analysis.

2. **Query genome.** The query genome is provided in 2bit format. Query-side annotations are not required for prediction but may be used for evaluation.

3. **Genome alignment chains.** Pairwise reference–query genome alignments are provided in UCSC chain format and define the search space used for locus projection. Low-scoring chains are filtered during loading.

2.3 Candidate region prediction

Input. Union exon models and genome alignment chains.

Operation. CURIA uses a chain-based locus-projection procedure adapted from the orthology-classification framework used in TOGA [1] to define coarse candidate regions in the query genome rather than reconstruct complete transcripts (Figure 2A).

For each chain–locus pair, seven features describing chain coverage and reference exon geometry are computed: log-transformed synteny gene count (`synteny_log1p`), global exonic occupancy (`gl_exo`), ±10 kb flank coverage (`flank_cov`), exonic-length-to-query-span ratio (`exlen_to_qlen`), exonic fraction (`exon_perc`), exon count (`ex_num`), and a single-exon indicator (`single_exon`).

A gradient-boosted decision-tree classifier, trained on labeled human–mouse chain–transcript pairs, assigns a correspondence probability to each pair. Pairs with predicted probability at least 0.5 are retained. For loci annotated as lncRNA, a relaxed threshold of 0.3 is used to favor candidate-region recall. These biotype-dependent thresholds are applied before the subsequent geometry-based branch routing.

For query genomes in which the classifier is overly conservative, candidate regions may instead be seeded from the highest-scoring alignment chain for each reference locus, irrespective of the classifier call.

For each retained chain–locus pair, the reference locus is projected through the corresponding alignment chain. The candidate query interval is defined by the span of chain blocks overlapping the reference locus and is extended by flanking sequence to accommodate local projection uncertainty. If multiple chains pass the selection criterion for the same reference locus, all corresponding candidate intervals are retained.

Output. Projected candidate query regions with their associated reference loci, chains, and classifier metadata.

Rationale. Downstream embedding-based comparison requires only a coarse locus-level search space. This avoids imposing coding-sequence constraints or attempting to reconstruct complete query transcripts.

2.3.1 Classifier training and evaluation partitioning

Chain–transcript pairs are labeled using the TOGA orthology-classification procedure on which this step is based [1]. Pairs assigned to the ORTH category form the positive class, whereas pairs assigned to the PARA category form the negative class; all other categories are excluded from training. These labels are used only to train the chain-based candidate-region classifier and should not be interpreted as ncRNA orthology assignments produced by CURIA.

The seven classifier features describe chain geometry and reference-side exon structure: synteny coverage, gene-level exonic overlap, flank coverage, exon-to-query-length ratio, exon fraction, exon count, and a single-exon indicator. Neither RNA embeddings nor query-side ncRNA annotations are used.

A gradient-boosted decision-tree classifier is trained on a balanced sample of up to 20,000 pairs per class. Model development uses a fixed random 75/25 stratified train/test split, and performance is summarized by five-fold stratified cross-validated AUC. Splits are stratified by class label but are not grouped by transcript, gene, locus, chromosome, or family. Related or neighboring loci may therefore occur in different folds, and the reported AUC should be interpreted as a measure of ORTH–PARA feature separability rather than as a strict held-out-locus generalization estimate.

The classifier is used only to define the candidate search space; downstream embedding-based matching and annotation-based evaluation are performed independently of its training labels.

2.4 RNA embeddings

Sequences are represented by per-nucleotide embeddings from the RiNALMo general-purpose RNA foundation model (`giga-v1`; 650M parameters, 1280-dimensional per-token embeddings) [25].

Two task-specific linear projections are used. A k=16 PCA, fitted on 10,368 per-token embeddings, is applied for position-level matching. A k=64 PCA, fitted on the 18,827 mean-pooled training windows described below, is applied for localized subregion detection.

The corresponding PCA transformations, the compact-ncRNA-versus-background detector, scan and match thresholds, and embedding settings are stored as versioned pipeline artifacts and selected through a model registry.

2.5 Short ncRNA pipeline

Compact single-exon ncRNA loci can be compared directly without decomposition into localized subregions (Figure 2C). This branch processes all single-exon loci of length up to 256 nt, irrespective of annotation biotype; typical examples include miRNAs, tRNAs, snoRNAs, and snRNAs, although short single-exon loci annotated as lncRNA may also enter this branch.

For each locus, the reference sequence is embedded, and the corresponding query region is symmetrically extended, when necessary, to provide a search span of 1.1 times the reference length. Candidate windows matching the reference length are tiled across the query region and compared with the reference using MMD (Section 2.6). The best-scoring window is then refined by local boundary perturbation of up to ±5 nt, subject to a final length constraint of 0.8–1.2× the reference length. Before comparison, embeddings are projected from 1280 to 16 dimensions using the matching PCA.

Output. Refined query coordinates with an MMD-based similarity score.

2.6 Similarity in embedding space (MMD)

For short ncRNAs, window similarity is quantified using Maximum Mean Discrepancy (MMD) [28] with an RBF kernel, treating per-position embeddings as empirical distributions:

$$\widehat{\mathrm{MMD}}_u^2(X, Y) = \frac{1}{n(n-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{nm} \sum_{i,j} k(x_i, y_j), \tag{1}$$

where $k(x, y) = \exp(-\gamma \|x - y\|^2)$ and $X = \{x_1, \ldots, x_n\}$ and $Y = \{y_1, \ldots, y_m\}$ are the sets of per-position embedding vectors for the two sequence windows. This is the unbiased empirical estimator, with self-terms excluded from the within-sample sums.

The implementation reports $\sqrt{\max(\widehat{\mathrm{MMD}}_u^2, 0)}$, because the unbiased finite-sample estimate may be negative. A reported value of zero may therefore result from clipping and does not necessarily indicate identical empirical distributions. Lower MMD values indicate more similar embedding distributions.

The kernel bandwidth γ is determined separately for each reference locus using the median pairwise-distance heuristic on its per-position embeddings and is then held fixed for all query-window comparisons at that locus. The resulting scores are suitable for ranking candidate windows within a locus, but are not globally calibrated across loci or biotypes. Cross-locus comparisons and global thresholds should therefore be interpreted approximately.

2.7 Reference island detection

To identify localized embedding patterns associated with annotated compact ncRNAs, we trained a logistic regression classifier on fixed-length genomic sequence windows represented in RiNALMo embedding space. Positive loci were sampled from annotated miRNAs, snoRNAs, snRNAs, scaRNAs, and `misc_RNAs`; lncRNAs were excluded from detector training.

For each selected single-exon ncRNA locus, the gene body together with 150 nt of flanking sequence on each side was divided into 128-nt windows with a stride of 24 nt. A window was labeled positive when it overlapped the annotated ncRNA by at least 20 nt. Other windows from the same region were labeled negative. Additional negative examples were generated by sampling N-free genomic regions of comparable length from the primary chromosomes and tiling them using the same window and stride.

Each window was embedded with RiNALMo, mean-pooled across positions, and projected from 1280 to 64 dimensions using the detection PCA before classifier training. Training loci were balanced across biotypes, with up to 150 loci per biotype and an additional 71 held-out loci per biotype used for threshold calibration. The committed training cache contains 18,827 windows: 6,538 positive and 12,289 negative. In five-fold cross-validation, the classifier achieved ROC-AUC ≈ 0.79. At a threshold calibrated to a background false-positive rate of at most 20%, it detected approximately 78% of windows overlapping an annotated element by at least 40 nt.

The detector therefore distinguishes embedding patterns associated with the compact ncRNA classes represented in its positive training set from the sampled genomic background. Contiguous detector-positive regions are termed "islands." This is an operational definition: detector positivity does not establish stable RNA structure, biological function, or membership in a general ncRNA class.

For reference-locus analysis, the concatenated exonic sequence of each union exon model is scanned using a 128-nt sliding window with a stride of 40 nt. Windows are embedded, mean-pooled, projected into the detection PCA space, and scored by the classifier. The resulting probability trace is smoothed with a five-window moving average, and contiguous regions exceeding a fixed probability threshold of 0.5 are defined as candidate islands.

The detection, projection, and embedding-based matching procedure is illustrated in Figure 3.

Island coordinates are subsequently mapped back to genomic space. An island that crosses a splice junction may therefore consist of multiple genomic segments. Single-exon loci of length 257–320 nt are not scanned but are represented as a single whole-exon island. This engineering heuristic avoids unstable decomposition of compact loci and does not imply end-to-end biological conservation.

Output. Reference islands with genomic coordinates and detector scores.

2.8 Chain-guided projection of islands

Input. Reference islands, retained chain–locus assignments, and genome alignment chains.

Reference islands are projected through the chain–locus assignments retained during candidate-region prediction (Figure 2A). In the standard projection mode, only assignments classified as ORTH are used. For the three marsupial comparisons (`monDom5`, `HLdidVir1`, and `HLnotEug3`), a relaxed best-chain mode was used: all ORTH assignments were retained, and for a reference locus without an ORTH assignment, the highest-scoring chain was added irrespective of its classifier category.

At most five highest-scoring chains per gene are retained, and chains are loaded with a minimum score of 25,000. Before projection, each reference island is extended by 144 nt on each side. Projections whose span exceeds 25 times the length of the flanked reference island are rejected. Each accepted query projection is then extended on both sides by the unflanked island length plus 256 nt.

Overlapping projected intervals on the same chromosome and strand are merged to avoid redundant scanning, while preserving links to all originating reference islands.

Output. Query search regions linked to their originating reference islands.

2.9 Local query island evaluation

Input. Merged query search regions, the query genome, and the detector described in Section 2.7.

Only query regions linked to at least one reference island are evaluated. Regions shorter than 128 bp or longer than 1.5 Mb are excluded. For regions longer than 2.5 kb, at least one linked reference locus must have a length of at least one quarter of the query-region length; this filter removes excessively expanded projections.

Each retained query region is scanned using the same window length, stride, smoothing procedure, and probability threshold as used for reference-island detection. Contiguous detector-positive windows are merged into query islands.

Output. Query islands with genomic coordinates and detector scores.

2.10 Reference–query island matching

2.10.1 Embedding-based local alignment

Each reference and query island is embedded as a complete sequence, and its per-token embeddings are projected from 1280 to 16 dimensions using the matching PCA. For a reference island with per-token matrix $A \in \mathbb{R}^{L_r \times 16}$ and a query island with matrix $B \in \mathbb{R}^{L_q \times 16}$, rows are L2-normalized and used to construct the per-token cosine-similarity matrix

$$C = \hat{A}\hat{B}^{\top},$$

where C_{ij} is the cosine similarity between reference position i and query position j.

Local alignment paths are extracted with a Smith–Waterman procedure [29] applied to the score matrix $S = C - \tau$, with cosine offset $\tau = 0.5$ and a linear gap penalty of 0.3. Let s denote the resulting embedding Smith–Waterman score. Match strength is reported as

$$d = \frac{1}{1+s},$$

where lower values indicate stronger matches.

Candidate matches are required to satisfy a minimum island length of 72 nt, a minimum effective aligned length of 40 nt, and a distance threshold of $d \leq 0.1$.

2.10.2 Match assignment and collinearity

Within each projected locus, candidate pairs are considered in order of increasing d. Each query island may be assigned to at most one reference island, whereas each reference island may retain at most two query islands, allowing a possible duplicated query copy while excluding further assignments.

After score-based assignment, matches are filtered for collinearity. For same-strand projections, query-island indices must be non-decreasing as reference-island indices increase; for opposite-strand projections, they must be non-increasing. The maximum-cardinality collinear subset is retained. When one reference island retains two query islands, the pairs are ordered by query-island index and their Smith–Waterman traceback start positions within the reference island are required to be non-decreasing. Assignment and collinearity filtering are performed independently for each locus.

Output. Scored reference–query subregion correspondences, together with their effective aligned lengths and genomic coordinates.

2.10.3 Interpretation of the embedding-SW score

Because the Smith–Waterman score accumulates cosine similarity along the alignment path, the derived distance $d = 1/(1+s)$ depends on both aligned length and per-position similarity. A longer alignment with moderate average similarity may therefore obtain a lower distance than a shorter alignment with higher average similarity. We consequently interpret d as an empirical, length-dependent ranking score rather than as a calibrated or length-normalized measure of similarity. Effective aligned length is recorded separately for each match.

2.10.4 Controlled matcher benchmark

To evaluate the local matcher under controlled conditions, we constructed an Rfam-derived benchmark (`notebooks/matching_benchmark.ipynb`). Positive pairs shared a curated Rfam seed-family core embedded in unrelated genomic flanks, whereas negative pairs contained cores from different families. We evaluated both discrimination between positive and negative pairs and localization of the planted homologous core. Numerical benchmark results are reported in the Results section.

2.10.5 Cross-locus and within-locus null analyses

Matched islands from the human–cow (`hg38-bosTau9`) and human–mouse (`hg38-mm39`) comparisons were evaluated using two complementary null analyses.

First, 250 matched island pairs were sampled at evenly spaced ranks from each complete match table. For each reference island, we recomputed embedding-SW distances to its assigned query partner, to a dinucleotide-preserving shuffle of that partner, and to the sampled query islands from other loci. The assigned partner was then ranked against the cross-locus candidate pool, and assigned-versus-cross-locus discrimination was summarized by ROC-AUC. Assigned pairs were also compared directly with their shuffled controls. For this null analysis, both query-sequence orientations were evaluated and the better score was retained. The production workflow instead extracts each query island in the strand orientation implied by the chain projection and evaluates that orientation only.

Second, 400 matched islands were sampled from each comparison for a within-locus positional analysis. Same-length windows were tiled across the projected query locus using a stride equal to half the assigned-island length, with a minimum stride of 25 nt. At most 100 evenly spaced windows were retained, and the assigned interval was always included. Windows whose midpoints lay within one assigned-island length of the assigned interval were excluded as overlapping alternatives. For loci with at least one remaining alternative, we recorded whether the assigned interval outscored all alternatives. For k alternatives, the corresponding locus-specific chance probability under exchangeability is $1/(k + 1)$.

These analyses evaluate cross-locus specificity, robustness to local nucleotide composition, and within-locus positional specificity. They do not establish biological function or evolutionary orthology.

2.11 Sequence-baseline and score-reducibility analyses

We tested whether embedding-based similarity was fully reducible to conventional nucleotide similarity by evaluating the compact-locus and localized-subregion workflows separately against the same panel of sequence-derived metrics: normalized Levenshtein identity, normalized affine-gap nucleotide Smith–Waterman score, aligned-region identity, 3-mer and 4-mer cosine similarity, dinucleotide-composition distance, length ratio, and GC-content difference.

For 2,994 human–mouse compact-locus predictions, we compared gene-grouped, five-fold cross-validated logistic regression models for any-exon annotation support. One model used only the sequence-derived metrics, whereas the second used the same metrics together with MMD.

For localized-subregion matching, 200 CURIA-assigned partners were sampled separately from the human–mouse and human–cow comparisons and contrasted with cross-locus alternatives. Models were evaluated using reference-island-grouped cross-validation and cluster-bootstrap confidence intervals. A prespecified low-sequence-similarity subset consisted of assigned pairs in the lowest quartile of normalized nucleotide Smith–Waterman score.

Within each syntenic locus, we additionally compared the rank of the CURIA-assigned partner with same-length tiled alternatives. All island-level AUC calculations used freshly recomputed embedding-SW scores for both assigned and alternative pairs. The stored match table was not used directly for these calculations because it had already been threshold-selected at $d \leq 0.1$ and therefore contained a restricted score range.

2.12 Evaluation metrics

Evaluation is based on agreement with existing annotations and is used as an approximate proxy for ground truth. Because ncRNA annotations are incomplete and heterogeneous across species, the reported metrics should be interpreted as measures of annotation agreement rather than direct estimates of biological accuracy.

2.12.1 Short ncRNA evaluation

For short ncRNAs, evaluation is performed at the locus level against annotated transcripts using three distinct criteria:

- *any-overlap rate*: the fraction of predictions intersecting an annotated ncRNA exon by at least 1 nt;
- 50% *prediction coverage*: the fraction for which the maximum exonic intersection with an annotated transcript, divided by the predicted-locus length, is at least 50%;
- *near-complete prediction coverage*: the fraction satisfying the same one-directional criterion at 99%.

We separately report workflow coverage across the full set of eligible reference loci: the number of non-protein-coding union loci satisfying the short-branch geometry, the number assigned at least one chain-projected query candidate, and the number producing a refined prediction. These counts describe processing yield rather than biological recall. Estimating recall would require a linked set of eligible human loci with established mouse counterparts; query-side interval overlap alone does not establish such a relationship.

As a conservative proxy, we also define a *named-counterpart* set consisting of eligible human loci whose gene name occurs exactly, ignoring case, in the mouse GENCODE annotation. A locus is counted as recovered when its final prediction overlaps an exon of the same-named mouse gene. This criterion is enriched for established, consistently named RNA families and should not be interpreted as a general genome-wide recall estimate.

2.12.2 Evaluation of localized-subregion predictions

For loci processed by the localized-subregion workflow, evaluation is performed at both the locus and subregion levels. A reference locus is considered supported when at least one predicted query subregion overlaps an annotated ncRNA exon. At the subregion level, a matched island is considered supported when it overlaps an annotated ncRNA exon in the query genome.

We report the fraction of supported loci, the fraction of matched islands overlapping annotated exons, the distribution of exonic overlap proportions, and the number of matched islands per locus. For multi-species analyses, recurrence is summarized by the number of query genomes in which a reference subregion has an accepted match.

To examine the relationship between embedding-based match strength and annotation support, predictions are stratified by score. We report annotation-overlap rates across score bins and compare score distributions between overlapping and non-overlapping predictions. Because the scores are not globally calibrated, these analyses characterize an empirical association with annotation support rather than a formal measure of prediction confidence.

Predictions outside annotated regions may represent either false positives or currently unannotated ncRNAs [10, 14, 15]. Annotation completeness varies across species, and ncRNA boundaries are often imprecise. Unlike protein-coding ORFs, many ncRNAs lack sharply defined, independently verifiable functional boundaries. Overlap-based evaluation should therefore be interpreted as an annotation-agreement measure rather than as a precise benchmark of biological accuracy or boundary recovery.

2.13 Recurrent-core aggregation and coding proximity

Reference islands obtained from different species comparisons are assigned to the same recurrent core when their human reference intervals overlap within the same gene. For each core and query species, only the accepted match with the lowest embedding-SW distance is retained.

A query species is counted as supporting a core when it contains an accepted island match. Because accepted matches already satisfy the pipeline threshold $d \leq 0.10$, the broad recurrent set is defined by support in at least 17 of 19 query species and by a mean reference-island length of at least 120 bp. This length is calculated across the retained species-specific human reference-island intervals contributing to the core, with at most one interval per query species; it is distinct from the total span of the aggregated core.

For sensitivity analyses, recurrence was recalculated using stricter per-species distance thresholds of $d \leq 0.05$, 0.04, 0.03, 0.02, and 0.01, while fixing the required species count at 17. We also varied the species quorum while fixing the per-species threshold at $d \leq 0.02$.

To characterize coding proximity, human lncRNA gene spans were compared with protein-coding gene spans. Direct overlaps were classified by strand as sense, antisense, or both. Non-overlapping lncRNA loci within 5 kb of a protein-coding gene were classified as near coding, and all remaining loci as intergenic. These categories describe genomic proximity only and do not imply transcriptional or functional relationships.

2.14 Exploratory ViennaRNA structure comparison

The exact human–mouse intervals for the three short-ncRNA examples in Figure 5 and the accepted MALAT1 recurrent-core matches R5, R4, R3, and R2 were analyzed with ViennaRNA 2.7.2 [30]. The similar-length short-ncRNA intervals were folded as complete sequences. For each MALAT1 core, length-matched windows were defined from the aligned span and expanded symmetrically by 20 or 40 nt.

Global partition-function folding and RNAplfold were applied. Human and mouse sequences were aligned using affine-gap nucleotide Smith–Waterman, and the resulting alignment was used to place both base-pair probability matrices in a shared coordinate system. Structural agreement was then summarized using a column-exact weighted Jaccard score and a ± 2-column tolerance-aware weighted overlap with one-to-one matching.

These metrics are exploratory descriptive measures and have not been validated as general structure-similarity scores. Full per-window and per-method results are provided in Supplementary Figures S1–S7, Supplementary Table S1, and the source TSV files.

2.15 Implementation details

Embeddings are computed with RiNALMo and projected using the two task-specific PCA transformations described in Section 2.4: $1280 \rightarrow 16$ for position-level matching and $1280 \rightarrow 64$ for localized-subregion detection. Compact-locus sequences are embedded separately for each candidate window, whereas complete reference and query islands are embedded to obtain the per-token matrices used by the local matcher.

Embedding inference is performed on GPU using batched execution. MMD computation, cosine-similarity alignment, match assignment, and downstream filtering are performed on CPU. CURIA is implemented as a staged task pipeline that separates embedding inference from subsequent analysis. Sequence windows are processed by a shared GPU worker using micro-batching, and intermediate representations are persisted for reuse by later stages and across query-genome runs.

RiNALMo inference uses PyTorch memory-efficient scaled-dot-product attention. Pipeline outputs are written in tabular and BED formats. Embedding-model versions, PCA transformations, detector parameters, and matching thresholds are stored as versioned pipeline artifacts.

3 Results

3.1 Benchmark setup

CURIA was evaluated on 19 human–query genome pairs, using human (`hg38`) as the reference and spanning evolutionary distances from primates to marsupials: rhesus macaque (`rheMac10`), mouse (`mm39`), rat (`rn7`), rabbit (`HLoryCun3`), cow (`bosTau9`), pig (`susScr11`), Bryde's whale (`HLbalEde1`), dromedary (`HLcamDro2`), horse (`equCab3`), cat (`felCat9`), dog (`canFam5`), pangolin (`HLmanJav2`), flying fox (`HLpteVam2`), hedgehog (`eriEur2`), armadillo (`dasNov3`), Asian elephant (`HLeleMax1`), gray short-tailed opossum (`monDom5`), Virginia opossum (`HLdidVir1`), and tammar wallaby (`HLnotEug3`).

Genome assemblies were obtained from NCBI [31], the UCSC Genome Browser [32], and DNA Zoo Consortium resources [33]. High-quality ncRNA annotations were used for human (GENCODE V48) and mouse (GENCODE VM37) [34]. The remaining query genomes were analyzed without relying on their native ncRNA annotations.

Evaluation followed the criteria defined in Section 2.12 and was performed at the locus level for compact ncRNA predictions and at the localized subregion level for island matches.

3.2 Overall performance

We first quantified processing yield for compact single-exon loci and agreement between refined query predictions and existing mouse annotations. Of 9,247 annotated non-protein-coding human union loci satisfying the single-exon, $\leq$ 256-nt routing criterion, 2,949 were assigned at least one chain-projected candidate region in the mouse genome, and 2,948 produced a refined prediction.

Because some reference loci were associated with multiple retained chains, these loci yielded 2,994 prediction rows. These rows form the denominator for the following annotation-overlap analysis; the preceding counts describe workflow coverage rather than recall against established orthologs. Among the 2,994 predictions, 52.4% overlapped an annotated exon by at least 1 nt, 46.5% had at least 50% of their length covered by an annotated exon, and 26.0% had at least 99% prediction coverage. These are one-directional prediction-coverage criteria rather than reciprocal-overlap measures. Among predictions with any annotated-exon overlap, the median covered fraction was 98.9% and the mean was 86.8%, indicating that annotation-supported predictions were generally well localized.

Under the conservative exact-name criterion, 446 geometry-eligible human loci had an annotated mouse gene with the same name. Predictions for 373 of these loci overlapped an exon of the corresponding same-named mouse gene, yielding an 83.6% named-counterpart recovery rate. Because this subset is enriched for established and consistently named RNA families, it should not be interpreted as genome-wide recall for short ncRNAs.

3.3 Short ncRNA performance and score behavior

We evaluated similarity between matched short ncRNA loci using MMD in embedding space. For the human–mouse pair (2,994 loci), the MMD distribution is skewed toward low values (median 0.142; Fig. 4A), with a substantial fraction of near-zero values (32.5% below 0.05), indicating consistent embedding representations between matched reference and query loci.

Stratification by biotype reveals clear differences (Fig. 4C). snoRNAs and tRNAs exhibit the lowest MMD values (median ≈ 0), consistent with their strong, well-characterized structural conservation [35]. miRNAs and lncRNAs occupy an intermediate range, whereas snRNAs, misc_RNAs, and rRNA-derived loci show higher and broader distributions. The rRNA category is too small and heterogeneous here for a biological conclusion about rRNA conservation; its distribution may instead reflect copy assignment, pseudogene contamination, projection, or boundary effects.

MMD is strongly associated with annotation overlap. Predictions with low MMD values ($\lesssim 0.1$) show high any-overlap rates with annotated loci ($\sim$88%), whereas higher MMD values show reduced overlap and increased dispersion. Predictions with MMD > 0.2 rarely overlap annotations and are enriched for ambiguous cases. Consistent with this, loci overlapping annotations are enriched at low MMD values, while non-overlapping predictions are shifted toward higher values (Fig. 4B), supporting the interpretation of MMD as a continuous empirical association with annotation support in this benchmark, rather than a globally calibrated confidence scale.

A subset of predictions with reported MMD values of zero did not overlap existing annotations. Because negative finite-sample estimates are clipped to zero, a reported zero should be interpreted as a very low estimated discrepancy rather than as evidence of identical embedding distributions.

Embedding similarity was strongly associated with nucleotide similarity but was not fully reducible to the tested sequence-derived metrics. Across all 2,994 short-ncRNA predictions, MMD was anticorrelated with nucleotide identity (Pearson $r = -0.80$; Spearman $\rho = -0.85$). A multivariate sequence-only model predicted annotation support with ROC-AUC 0.887, compared with 0.893 after adding MMD (ΔROC-AUC = +0.0064, 95% CI $[0.0030, 0.0097]$). This small increment disappeared when miRNAs were excluded ($\Delta = -0.0007$), indicating that MMD adds little predictive information over the full sequence baseline for the remaining short-ncRNA classes in this benchmark.

3.4 Localized embedding-supported regions ("islands")

We analyzed localized subregions ("islands") within annotated human lncRNA loci to assess cross-species correspondence below the locus level. Reference–query island matching uses Smith–Waterman local alignment over the per-token cosine similarity matrix, and match strength is reported as an embedding-SW distance $d = 1/(1 + \text{score})$, with lower values indicating stronger matches (Section 2.10).

Of 34,846 annotated human lncRNA union loci, 229 met the compact single-exon routing criterion and were processed by the compact-locus workflow rather than reference-island detection. Of the remaining 34,617 loci, 34,532 had a retained chain–locus assignment that projected successfully in at least one of the 19 comparisons and therefore entered reference-island processing; the other 85 had no successful candidate-region projection in any comparison.

Among the 34,532 processed loci, 18,364 (53.2%) contained at least one reference island. Island-positive loci contained a mean of 1.51 reference islands and a median of 1, yielding 27,784 detected islands in total (Fig. 6A). Because islands may cross splice junctions, these intervals corresponded to 43,506 genomic segments. Thus, when detector-positive signal was present, it was generally concentrated in a small number of localized regions within the union exon model.

Across accepted matches in the multi-species panel, embedding-SW distance was moderately anticorrelated with effective aligned length (Spearman $\rho \approx -0.64$), and the median effective aligned length was approximately 150 nt. Matches below the median aligned length had a median distance of approximately 0.06, compared with approximately 0.03 for matches at or above the median. This length dependence confirms that d reflects both alignment extent and per-position similarity and should not be interpreted as a calibrated or length-normalized similarity measure (Section 2.10).

In the controlled Rfam-derived matcher benchmark, the embedding-SW score distinguished planted same-family pairs from different-family pairs with ROC-AUC 0.993 and localized the planted core in all 140 positive pairs. This benchmark is a controlled sanity check of the local matcher rather than an estimate of genome-wide annotation accuracy (Section 2.10).

In the human–mouse comparison, 50.1% of accepted query-island matches overlapped an annotated mouse ncRNA exon by at least 1 nt, and 28.6% had at least 50% of their length covered by annotated exons.

Together, these observations are consistent with a modular pattern of cross-species correspondence, in which detectable similarity is concentrated in localized regions rather than distributed uniformly across the annotated locus [10–12, 14, 36]. This interpretation is also compatible with well-characterized lncRNAs such as NEAT1, XIST, and MALAT1, in which distinct localized domains contribute to different molecular functions [37–39].

3.4.1 Multi-species recurrent cores

Here, a "core" is defined operationally as a human reference-coordinate region with accepted localized-subregion matches in one or more query genomes. Species-specific reference islands are assigned to the same core when their human coordinates overlap within the same gene.

Across all biotypes processed by the localized-subregion workflow, aggregation of accepted matches from the 19 comparisons produced 31,977 reference-coordinate cores with at least one accepted match (Fig. 6B–C). This is a conditional denominator that excludes detected reference islands for which no accepted query match was obtained. Among these cores, 87% had accepted matches in at least two query genomes, 67% in at least five, and approximately 2% in all 19 query genomes, including the three marsupial genomes. Absence of a species may reflect either failure to obtain a projected candidate region or failure to identify an accepted match within that region.

The 31,977 aggregated cores and the 27,784 reference detector islands reported above are not directly comparable counts. The latter are unique islands detected in human lncRNA union exon models, whereas the former are reconstructed from accepted match rows across all species comparisons and all biotypes routed to the localized-subregion workflow. Species-specific reference intervals are merged into cores by overlap in human reference coordinates.

We next tested whether accepted matches could be explained by cross-locus similarity, local nucleotide composition, or selection of the best among many positions within a projected locus. In the cross-locus analysis, the assigned query partner ranked within the top 5% of the sampled candidate pool for 69% of human–cow and 65% of human–mouse reference islands. The corresponding assigned-versus-cross-locus ROC-AUC values were 0.907 and 0.898. Assigned pairs also had lower embedding-SW distances than dinucleotide-preserving shuffles of their query sequences in 94% of human–cow and 92% of human–mouse cases.

The within-locus positional analysis included 287 human–cow and 308 human–mouse loci with at least one non-overlapping alternative window. The assigned interval outscored all alternatives in 189 of 287 cow loci (65.9%) and 202 of 308 mouse loci (65.6%), compared with mean locus-specific chance probabilities of 15.6% and 15.9%, respectively. Among loci containing at least 20 alternatives, the assigned interval ranked first in 50% of cow loci and 58% of mouse loci, compared with a chance expectation of approximately 3%. Nevertheless, the assigned interval did not rank first in approximately one third of evaluable loci, indicating substantial residual positional ambiguity.

Together, these controls argue against purely cross-locus, composition-driven, or simple best-of-many positional explanations for the observed matches in the tested human–mouse and human–cow samples. They do not establish biological function or evolutionary orthology for individual cores.

A complementary score-reducibility analysis (Section 2.11) tested whether embedding-SW provided ranking information beyond nucleotide identity, local sequence alignment, k-mer similarity, nucleotide composition, and length. Adding embedding-SW to the multivariate sequence-only baseline increased ROC-AUC from 0.897 to 0.935 for human–mouse ($\Delta = +0.038$, 95% cluster-bootstrap CI $[0.022, 0.056]$) and from 0.912 to 0.952 for human–cow ($\Delta = +0.040$, 95% cluster-bootstrap CI $[0.026, 0.055]$).

Within the prespecified low-sequence-similarity subset, embedding-SW showed greater discrimination than nucleotide Smith–Waterman: ROC-AUC was 0.798 versus 0.615 for human–mouse and 0.796 versus 0.616 for human–cow. Within the same projected locus, embedding-SW also ranked the CURIA-assigned partner first more often than nucleotide Smith–Waterman (31.0% versus 18.4% for human–mouse and 24.4% versus 11.6% for human–cow). This analysis measures reproduction of the pipeline's assignment rather than identification of a known biological counterpart.

3.5 Broadly recurrent cores across the species panel and coding proximity

Human lncRNA annotation contains tens of thousands of loci, but the degree of biological and mechanistic characterization varies widely [13, 14]. We therefore asked whether recurrence across a broad species panel could define smaller sets of loci for further investigation.

The near-panel-wide recurrent set was obtained through a defined filtering funnel. The reference annotation contained 34,846 human lncRNA union loci. Of these, 229 satisfied the compact single-exon routing criterion and were processed by the compact-locus workflow, while 85 lacked a successful candidate-region projection in all 19 comparisons. The remaining 34,532 loci entered the localized-subregion workflow.

Among these loci, 18,364 contained at least one detected reference island, and 11,328 were projectable in at least 17 query genomes. Requiring the same reference-coordinate core to have an accepted match in at least 17 genomes retained 1,761 annotated lncRNA loci. Applying the additional requirement that the mean length of the retained species-specific reference-island intervals was at least 120 bp yielded a broad recurrent set of 1,693 loci (Table 1).

Accepted per-species matches already satisfy the pipeline threshold $d \leq 0.10$; an additional cutoff on the mean distance across species therefore does not impose substantial independent stringency. Final locus counts are aggregated by version-stripped Ensembl gene identifier, so the 1,693 entries represent annotated human lncRNA gene loci rather than individual islands or newly inferred lncRNA annotations.

The 17-of-19 recurrence criterion permits two biological or technical non-detections. Failure to recover a core in an individual genome may reflect projection failure, insufficient embedding similarity, assembly limitations, or true evolutionary divergence and is not interpreted by itself as lineage-specific absence.

Of the 1,693 loci, 379 were intergenic relative to human protein-coding gene annotations, defined as neither overlapping nor lying within 5 kb of a protein-coding gene. The presence of this intergenic subset shows that the recurrent set is not restricted to loci directly overlapping protein-coding genes. However, conservation of other DNA-level regulatory or repetitive elements remains a possible confounding source of signal.

Because the broad recurrent set is defined primarily by species coverage and the pipeline-wide acceptance threshold, we repeated the 17-of-19 analysis using stricter per-species distance requirements. Requiring every counted species to satisfy $d \leq 0.05$, 0.03, and 0.02 retained 784, 379, and 170 loci, respectively, of which 187, 103, and 42 were intergenic. A threshold of $d \leq 0.01$ retained 21 loci. These nested sets show that recurrent matches are not confined to scores near

Table 1: Filtering funnel from annotated human lncRNA gene loci to the broad recurrent-core set.

Stage	Unit	Count
Annotated human lncRNA union loci	gene locus	34,846
Entered localized-subregion processing in at least one comparison	gene locus	34,532
Contained at least one detected reference island	gene locus	18,364
Projectable in at least 17 query genomes	gene locus	11,328
Contained at least one core matched in at least 17 genomes	gene locus	1,761
After mean reference-island length $\geq 120\,\mathrm{bp}$	gene locus	1,693
Intergenic relative to protein-coding genes ($> 5\,\mathrm{kb}$)	gene locus	379

Table 2: Sensitivity of recurrent-core counts to the per-species embedding-SW distance cutoff at a fixed quorum of at least 17 of 19 query genomes.

Embedding-SW cutoff	Required species	Cores	lncRNA loci	% broad set	Intergenic loci
$d \leq 0.10$	≥ 17	1,813	1,693	100.0	379
$d \leq 0.05$	≥ 17	837	784	46.3	187
$d \leq 0.03$	≥ 17	404	379	22.4	103
$d \leq 0.02$	≥ 17	173	170	10.0	42
$d \leq 0.01$	≥ 17	21	21	1.2	6

the pipeline acceptance threshold, while also demonstrating that the broad recurrent set is substantially larger than the more stringently filtered subsets.

Core identity is anchored to human reference coordinates. Species-specific reference intervals that overlap within the same gene are assigned to the same core, whereas non-overlapping intervals remain separate. Among cores passing the 17-of-19 screen, none were formed by transitively merging pairwise-disjoint reference intervals.

3.6 Case studies

3.6.1 Compact ncRNA projections

To illustrate the behavior of CURIA, we examined three representative compact ncRNA loci (Fig. 5). Projection of human SNORD57 to mouse identified a match localized over the annotated *Snord57* locus, illustrating recovery of a strongly conserved compact ncRNA. The human RNU6-7 projection intersected the annotated mouse locus *Gm24859*. For human vault RNA (ENSG00000199990), CURIA identified a corresponding region overlapping *Vaultrc5*; although the predicted mouse interval differed in length from the human transcript, the method localized the best-matching segment within the syntenic search region.

The SNORD57 and RNU6-7 predictions overlapped their respective mouse annotations with edit-derived identities of 74% and 93%, and both had reported MMD values of 0.00. The vault-RNA prediction overlapped *Vaultrc5* despite lower sequence similarity (59% edit-derived identity; normalized nucleotide-SW score 0.35; MMD 0.14). These examples span a range of primary-sequence conservation and were examined further using exploratory ViennaRNA-based structure comparisons.

3.6.2 Localized correspondence in long ncRNA loci

Inspection of representative long ncRNA loci revealed distinct patterns of localized cross-species correspondence. MALAT1 showed the clearest recurrent pattern: the same terminal reference cores produced strong accepted matches in mouse, rat, cat, cow, armadillo, and opossum, including genomes in which no native MALAT1 annotation was available. XIST showed a different but similarly localized pattern, with several recurrent cores clustering near the transcript 5' region in mouse, rat, cat, cow, and armadillo, while internal matches were less consistently recovered. NEAT1 exhibited a more heterogeneous modular pattern, with several recurrent terminal and internal cores interspersed among regions that were inconsistently matched across species. NORAD was dominated by a single broadly recurrent core across most of the species panel. By contrast, FIRRE illustrated a projection-limited case in which extensive locus divergence and fragmented chain coverage restricted downstream matching.

TERC was strongly recovered in rat, cat, cow, and armadillo. In mouse, CURIA selected a candidate locus overlapping the annotated locus *Gm55883*, where a robust human–TERC match was recovered. A second chain projected the human locus directly onto the annotated mouse *Terc* gene, but covered essentially only the gene itself, without appreciable flanking or broader locus-level support, and was therefore rejected by the chain-based candidate classifier. The discrepancy between the selected syntenic locus and the current mouse *Terc* annotation was not resolved here.

These locus-level observations are qualitative examples of the different correspondence patterns recovered by CURIA and do not by themselves establish conserved molecular function or evolutionary orthology. Representative stacked genome-browser views for MALAT1 and XIST are provided in Supplementary Figs. S8 and S9, respectively.

3.7 Exploratory comparison with predicted secondary structure

We compared the three representative compact ncRNA pairs and four accepted MALAT1 recurrent-core matches using alignment-aware ViennaRNA predictions. Base-pair probability matrices for the human and mouse sequences were mapped into shared nucleotide-alignment coordinates before comparison.

RNU6-7, included as a positive control, showed nearly identical predicted pairing, whereas SNORD57 showed partial structural agreement. Both vault-RNA sequences contained prominent predicted long-range pairing, but their alignment-aware base-pair contacts showed little correspondence. Among the MALAT1 examples, R5 showed the strongest and most stable moderate agreement across the tested window boundaries and folding procedures. R4 showed weak, boundary-dependent agreement; R3 showed weak agreement that differed between global and local folding; and R2 remained consistently near zero across the tested windows (Supplementary Figures S1–S7 and Supplementary Table S1).

These heterogeneous results show that strong embedding-based correspondence is not uniformly accompanied by similar ViennaRNA-predicted base-pairing patterns. The analysis is exploratory and does not establish structural conservation, biological function, or evolutionary orthology.

3.8 Observed limitations and failure modes

Recovery depends first on successful chain-based projection of the corresponding query region. Loci absent from the available genome alignments, or affected by assembly gaps, fragmentation, and local sequence errors, may therefore be missed or only partially projected. Incomplete or imprecise reference annotations can further limit the recoverable signal, particularly for long and structurally complex ncRNA loci.

Among successfully projected loci, some union exon models contain weak, diffuse, or fragmented detector-positive signal and consequently yield no reference islands or no accepted reference–query matches. These cases can reflect genuine divergence, incomplete assemblies or annotations, insufficient detector sensitivity, or limitations of the embedding-based matcher. The current evaluation does not systematically distinguish among these causes.

3.9 Runtime and scalability

Across the 19 genome-pair panel, a complete reference–query run required approximately three hours on a single NVIDIA RTX PRO 4500 GPU under the benchmark configuration used here. Embedding inference was performed on GPU using batched execution, whereas MMD computation, local matching, assignment, and filtering were performed on CPU. Intermediate reference-side results were persisted and reused across query-genome runs.

4 Discussion

CURIA should be viewed primarily as a proof of concept showing that synteny-constrained embedding comparison can recover informative cross-species ncRNA correspondences at genome scale, rather than as a production-ready annotation framework optimized for exhaustive sensitivity. Within this framing, several main observations emerge.

4.1 Key findings

This study provides a genome-scale proof of concept for identifying candidate cross-species ncRNA correspondences by combining syntenic constraints with embedding-based similarity. The approach recovered annotation-supported loci and localized matches across examples with varying levels of primary-sequence similarity.

For compact single-exon ncRNAs, MMD provided an empirical refinement and ranking score. The more distinctive result arose from localized-subregion matching: in the tested human–mouse and human–cow samples, embedding-SW added ranking information beyond nucleotide alignment, identity, k-mer similarity, composition, and length.

Accepted correspondences were often restricted to localized subregions rather than extending across complete annotated loci. Together, these findings are consistent with heterogeneous and modular cross-species ncRNA correspondence and show that localized embedding-space comparison can provide information complementary to conventional sequence-based measures.

4.2 Comparison to existing methods

Existing methods address complementary aspects of ncRNA discovery, homology, and structural comparison. Covariance-model approaches identify members of curated RNA families, whereas structure-aware alignment methods compare already defined RNA sequences. Comparative genome-wide screens can detect candidate conserved RNA elements but are not designed primarily for reference-guided transfer of individual ncRNA annotations.

Methods developed specifically for lncRNA evolution are more directly comparable. slncky combines transcript discovery with whole-genome-alignment-based analysis of sequence conservation and syntenic correspondence [40]. lncHOME identifies lncRNAs with conserved genomic positions and related patterns of RNA-binding-protein motifs, and uses these features to define candidate homologous families across vertebrates [41]. These approaches have demonstrated that subsets of lncRNAs can be traced across substantial evolutionary distances using sequence-, position-, expression-, or motif-based evidence.

CURIA addresses a related but distinct setting. It starts from a reference annotation, a query genome, and pairwise genome alignment chains, without requiring a query-side transcript annotation, a curated RNA-family model, or predefined RNA-binding motifs. Within syntenically projected loci, it localizes and ranks candidate corresponding subregions using RNA-language-model representations. This design is intended for loci in which correspondence may be restricted to localized regions and conventional sequence-level evidence is weak, heterogeneous, or incomplete.

The present results do not establish that embedding-based matching supersedes existing homology methods. Rather, they indicate that localized embedding similarity can provide an additional comparative signal that may be integrated with sequence conservation, transcript evidence, structural models, and functional genomic data.

4.3 Biological implications

For loci processed by the localized-subregion workflow, accepted cross-species correspondences were often confined to compact regions within broader annotated loci. This pattern is consistent with the view that evolutionarily retained information in lncRNAs may be concentrated in localized elements rather than distributed uniformly across the complete transcript [10, 12, 14].

Embedding-based similarity was correlated with conventional sequence similarity, but the relationship was not one-to-one. A subset of candidate correspondences showed low nucleotide identity despite strong embedding similarity, indicating that the two measures can rank the same pairs differently.

The localized-subregion representation provides a practical way to capture this heterogeneity. Recurrently matched islands define candidate corresponding regions, whereas other parts of the same annotated locus may show no detectable localized match under the current model. Correspondence patterns also differ among RNA classes: compact RNAs such as tRNAs and snoRNAs generally show stronger and more consistent signal, whereas lncRNA loci more often yield partial or fragmented matches.

Given the heterogeneity of lncRNA evolution and annotation, a single universal orthology criterion may be premature. A more tractable near-term objective is to stratify candidate correspondences according to the strength and type of comparative evidence. Recurrently matched candidate cores provide one such evidence layer and define subsets that can be prioritized for subsequent functional and evolutionary investigation, without treating recurrence alone as evidence of orthology or function.

4.4 Limitations

The principal limitation is the absence of reliable ground truth for cross-species ncRNA correspondence. Existing annotations are incomplete and heterogeneous across species, and transcript boundaries—particularly for lncRNAs—are often ambiguous. Annotation overlap therefore provides an approximate measure of agreement rather than a direct estimate of biological accuracy.

The recurrent-core analysis also separates species coverage from match stringency. At the pipeline acceptance threshold of $d \leq 0.10$, the 17-of-19 set contains 1,693 lncRNA loci, whereas requiring every counted species to satisfy $d \leq 0.02$ or $d \leq 0.01$ retains 170 and 21 loci, respectively. Because embedding-SW distance depends on aligned length as well as per-position similarity, these nested sets should be interpreted as sensitivity analyses rather than calibrated biological confidence classes.

CURIA also depends on the availability and quality of whole-genome alignment chains. Corresponding regions outside chain coverage cannot be recovered, and assembly gaps, fragmentation, or alignment errors can propagate into missed or distorted candidate regions. The candidate-region classifier is adapted from the chain-based orthology framework used in TOGA [1]. Although its thresholds and routing were adjusted for ncRNA analysis, its features retain assumptions derived from protein-coding gene projection and may not capture the full diversity of ncRNA locus evolution.

The method further depends on the representation learned by the RNA foundation model and on the detector trained in that embedding space. The current detector was trained to distinguish windows associated with annotated compact ncRNA classes from sampled genomic background. A genuine lncRNA element whose representation differs from those training classes may therefore receive a low detector score and never reach the local matcher. Held-out-family analyses showed broader generalization among structured-RNA families than for lncRNAs, consistent with this limitation.

More generally, CURIA is sensitive to localized similarity represented by the selected embedding model. It does not explicitly model long-range interactions, global RNA architecture, or rearrangement of functional elements within a locus. The framework itself is not tied to RiNALMo: alternative embedding models, detectors, and matching functions can be substituted, but their behavior and boundary sensitivity would require separate calibration and evaluation.

Predicted regions do not establish transcriptional activity, molecular function, or evolutionary orthology. An annotation-discordant match may reflect incomplete query annotation, projection or assembly error, correspondence to a nearby element, or a false-positive assignment; interval overlap alone cannot distinguish among these possibilities. Recurrent localized-subregion matches may also reflect conserved sequence features that are not specific to RNA structure or function.

Finally, CURIA does not explicitly distinguish shared evolutionary origin from convergent similarity in embedding space. Syntenic restriction favors loci with shared genomic ancestry, but embedding similarity may also arise from convergent structural features or recurrent local motifs. Accepted matches should therefore be interpreted as candidate corresponding elements supported jointly by synteny and embedding similarity, rather than as definitive evidence of orthology, structure, or function.

4.5 Embedding-based similarity and its interpretation

Embedding similarity retained a strong association with primary-sequence similarity, but the two measures did not rank all candidate correspondences identically. For compact single-exon ncRNAs, MMD largely tracked the tested sequence-derived metrics. In contrast, localized embedding-SW matching provided additional ranking information in the human–mouse and human–cow analyses beyond nucleotide identity, local sequence alignment, k-mer similarity, nucleotide composition, and length.

The embedding-based detector provides a second use of the learned representation. It distinguishes windows associated with the compact ncRNA classes represented in its training set from sampled genomic background and thereby identifies localized candidate regions for subsequent matching. Detector positivity is an operational property of the learned representation and does not by itself imply RNA structure, expression, or function.

Previous work has shown that RNA foundation-model representations can encode secondary-structure-related and functional properties [24, 26, 27, 42]. The present analyses do not determine which learned properties account for the additional ranking information observed in the localized-subregion workflow. ViennaRNA-predicted structural agreement was heterogeneous across the examined examples, ranging from moderate and boundary-stable agreement to consistently weak agreement. Embedding-supported correspondence should therefore not be interpreted as a direct measure of conserved secondary structure.

Taken together, these results indicate that, within syntenically constrained loci, learned RNA representations provide a localized comparison signal that is not fully reproduced by the tested conventional sequence metrics. This signal may reflect combinations of sequence context, nucleotide composition, local motifs, structural propensities, and other properties learned by the model, but the current analysis does not assign it to any single biological mechanism.

4.6 Future directions

The immediate next step is to examine the recurrent cores in greater biological detail. Candidate regions can be prioritized using transcriptomic support, predicted secondary structure, RNA–protein interaction data, evolutionary constraint, and targeted experimental validation. Systematic comparison of core number, order, duplication, and divergence across species may reveal gain, loss, and rearrangement of localized ncRNA elements that are difficult to detect at whole-transcript scale.

The candidate-locus stage could also be expanded beyond the current chain-classification procedure. More flexible projection models could combine whole-genome alignments with local sequence evidence, neighboring gene context, transcriptomic data, and multiple candidate chains. Such models may improve sensitivity for rapidly evolving loci, duplicated regions, and assemblies in which a single best chain does not adequately represent the corresponding search space.

Localized-subregion detection is another important target for improvement. The current detector is trained primarily on compact annotated ncRNA classes and may therefore favor embedding patterns resembling those examples. Broader training sets, class-specific detectors, multi-scale windows, and models trained directly on recurrent cross-species signal could improve sensitivity to lncRNA-specific elements and reduce dependence on features characteristic of compact ncRNAs. Larger RNA foundation models, or models with independently demonstrated structural or functional information in their representations, may also provide more informative embeddings for both detection and matching.

The local matcher is likewise not assumed to be optimal. Smith–Waterman alignment over the token-level cosine-similarity matrix provides a simple order-aware proof-of-concept objective, but its accumulated score depends on both alignment length and per-position similarity. Alternatives include length-normalized or length-regularized local alignment, soft-DTW and related order-aware objectives, optimal-transport-based matching, and local substring search based on MMD or other distributional distances with explicit constraints against degenerate short matches. Different RNA classes may ultimately require different detection and matching strategies.

Extending the analysis from tens to hundreds of genomes would enable a more explicit evolutionary treatment of recurrent cores. Phylogenetic models could separate broadly conserved ancestry from recurrent similarity, reduce sensitivity to individual assemblies or chain sets, and identify lineage-specific gains, losses, duplications, and rearrangements. Once the correspondence framework is better calibrated, it could be used to ask why a candidate core is retained across most of a clade but absent from a particular lineage, or to identify elements restricted to primates, apes, or other evolutionary groups. These directions would shift CURIA from a proof-of-concept correspondence framework toward a comparative platform for studying the evolution, organization, and lineage specificity of ncRNA loci.

5 Data and code availability

The CURIA pipeline, analysis code, and documentation are available at `https://github.com/kirilenkobm/CURIA_pipeline`. The complete per-species output snapshot used for this study is archived on Zenodo [43] under DOI `10.5281/zenodo.21383175`. The archive contains the full `hg38_vs_*` result directories for all 19 query assemblies, including candidate-region classifications, reference and query island coordinates, accepted localized matches, compact-ncRNA predictions, union-transcript mappings, and run metadata. The repository directory `preprint_results/` contains a README and a download script that retrieve this exact archived snapshot and verify its SHA-256 checksum.

The recurrent-core tables described in Section 3 are included in the repository under `paper/recurrent_core_tables/`. This directory contains gene-level, core-level, and core-by-assembly match tables, distance and quorum sensitivity tables, and a README defining coordinates, identifiers, acceptance criteria, and derived fields.

The complete supplementary material is provided as `paper/supplementary/supplementary.pdf`. It contains Supplementary Figures S1–S9 and Supplementary Table S1, including the exploratory ViennaRNA-based comparisons and representative locus-level browser views for MALAT1 and XIST. Machine-readable per-window results, source data, and diagnostic plots for the ViennaRNA analysis are provided under `analysis/vienna_structure_examples/`, including `malat1_windows.tsv`.

Genome assemblies in 2bit format were obtained from NCBI [31], the UCSC Genome Browser [32], and DNA Zoo Consortium resources [33]. Genome-alignment chains were obtained from the TOGA project (Hiller lab) [1]: `https://genome.senckenberg.de/download/TOGA`. All external datasets are publicly available, and links to the exact sources are provided in the repository.

References

[1] Bogdan M. Kirilenko, Chetan Munegowda, Ekaterina Osipova et al. Integrating gene annotation with orthology inference at scale. *Science*, 380(6643):eabn3107, April 2023. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.abn3107.

[2] Virag Sharma and Michael Hiller. Increased alignment sensitivity improves the usage of genome alignments for comparative gene annotation. *Nucleic Acids Research*, 45(14): 8369–8377, August 2017. ISSN 0305-1048, 1362-4962. doi: 10.1093/nar/gkx554.

[3] W. James Kent, Robert Baertsch, Angie Hinrichs et al. Evolution's cauldron: Duplication, deletion, and rearrangement in the mouse and human genomes. *Proceedings of the National Academy of Sciences*, 100(20):11484–11489, September 2003. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1932072100.

[4] Virag Sharma, Peter Schwede, and Michael Hiller. CESAR 2.0 substantially improves speed and accuracy of comparative gene annotation. *Bioinformatics*, 33(24):3985–3987, December 2017. ISSN 1367-4803, 1367-4811. doi: 10.1093/bioinformatics/btx527.

[5] Javier Herrero, Matthieu Muffato, Kathryn Beal et al. Ensembl comparative genomics resources. *Database*, 2016:bav096, 2016. ISSN 1758-0463. doi: 10.1093/database/bav096.

[6] Walter M. Fitch. Distinguishing Homologous from Analogous Proteins. *Systematic Zoology*, 19(2):99, June 1970. ISSN 00397989. doi: 10.2307/2412448.

[7] Eugene V. Koonin. Orthologs, Paralogs, and Evolutionary Genomics. *Annual Review of Genetics*, 39(1):309–338, December 2005. ISSN 0066-4197, 1545-2948. doi: 10.1146/annurev.genet.39.073003.114725.

[8] Igor Ulitsky and David P. Bartel. lincRNAs: Genomics, Evolution, and Mechanisms. *Cell*, 154(1):26–46, July 2013. ISSN 00928674. doi: 10.1016/j.cell.2013.06.020.

[9] Sven Diederichs. The four dimensions of noncoding RNA conservation. *Trends in Genetics*, 30(4):121–123, April 2014. ISSN 01689525. doi: 10.1016/j.tig.2014.01.004.

[10] Hadas Hezroni, David Koppstein, Matthew G. Schwartz et al. Principles of Long Noncoding RNA Evolution Derived from Direct Comparison of Transcriptomes in 17 Species. *Cell Reports*, 11(7):1110–1122, May 2015. ISSN 22111247. doi: 10.1016/j.celrep.2015.04.023.

[11] Elena Rivas. Evolutionary conservation of RNA sequence and structure. *WIREs RNA*, 12 (5):e1649, September 2021. ISSN 1757-7004, 1757-7012. doi: 10.1002/wrna.1649.

[12] Igor Ulitsky. Evolution to the rescue: Using comparative genomics to understand long non-coding RNAs. *Nature Reviews Genetics*, 17(10):601–614, October 2016. ISSN 1471-0056, 1471-0064. doi: 10.1038/nrg.2016.85.

[13] Luisa Statello, Chun-Jie Guo, Ling-Ling Chen et al. Gene regulation by long non-coding RNAs and its biological functions. *Nature Reviews Molecular Cell Biology*, 22(2):96–118, February 2021. ISSN 1471-0072, 1471-0080. doi: 10.1038/s41580-020-00315-9.

[14] John S. Mattick, Paulo P. Amaral, Piero Carninci et al. Long non-coding RNAs: Definitions, functions, challenges and recommendations. *Nature Reviews Molecular Cell Biology*, 24(6): 430–447, June 2023. ISSN 1471-0072, 1471-0080. doi: 10.1038/s41580-022-00566-8.

[15] Mihaela Pertea, Alaina Shumate, Geo Pertea et al. CHESS: A new human gene catalog curated from thousands of large-scale RNA sequencing experiments reveals extensive transcriptional noise. *Genome Biology*, 19(1):208, December 2018. ISSN 1474-760X. doi: 10.1186/s13059-018-1590-2.

[16] Eric P. Nawrocki and Sean R. Eddy. Infernal 1.1: 100-fold faster RNA homology searches. *Bioinformatics*, 29(22):2933–2935, November 2013. ISSN 1367-4811, 1367-4803. doi: 10.1093/bioinformatics/btt509.

[17] Nancy Ontiveros-Palacios, Emma Cooke, Eric P Nawrocki et al. Rfam 15: RNA families database in 2025. *Nucleic Acids Research*, 53(D1):D258–D267, January 2025. ISSN 0305-1048, 1362-4962. doi: 10.1093/nar/gkae1023.

[18] Sebastian Will. LocARNA 2.0: Versatile Simultaneous Alignment and Folding of RNAs. In Ronny Lorenz, editor, *RNA Folding*, volume 2726, pages 235–254. Springer US, New York, NY, 2024. ISBN 978-1-0716-3518-6 978-1-0716-3519-3. doi: 10.1007/978-1-0716-3519-3_10.

[19] Matthias Höchsmann, Thomas Töller, Robert Giegerich et al. Local similarity in RNA secondary structures. *Proceedings. IEEE Computer Society Bioinformatics Conference*, 2: 159–168, 2003. ISSN 1555-3930.

[20] Andreas R. Gruber, Sven Findeiß, Stefan Washietl et al. RNAz 2.0: Improved noncoding RNA detection. *Pacific Symposium on Biocomputing. Pacific Symposium on Biocomputing*, pages 69–79, 2010. ISSN 2335-6936.

[21] Jakob Skou Pedersen, Gill Bejerano, Adam Siepel et al. Identification and Classification of Conserved RNA Secondary Structures in the Human Genome. *PLoS Computational Biology*, 2(4):e33, April 2006. ISSN 1553-7358. doi: 10.1371/journal.pcbi.0020033.

[22] Zoonomia Consortium, Diane P. Genereux, Aitor Serres et al. A comparative genomics multitool for scientific discovery and conservation. *Nature*, 587(7833):240–245, November 2020. ISSN 0028-0836, 1476-4687. doi: 10.1038/s41586-020-2876-6.

[23] Joel Armstrong, Glenn Hickey, Mark Diekhans et al. Progressive Cactus is a multiple-genome aligner for the thousand-genome era. *Nature*, 587(7833):246–251, November 2020. ISSN 0028-0836, 1476-4687. doi: 10.1038/s41586-020-2871-y.

[24] Jiayang Chen, Zhihang Hu, Siqi Sun et al. Interpretable RNA Foundation Model from Unannotated Data for Highly Accurate RNA Structure and Function Predictions, August 2022.

[25] Rafael Josip Penić, Tin Vlašić, Roland G. Huber et al. RiNALMo: General-purpose RNA language models can generalize well on structure prediction tasks. *Nature Communications*, 16(1):5671, July 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-60872-5.

[26] Manato Akiyama and Yasubumi Sakakibara. Informative RNA base embedding for RNA structural alignment and clustering by deep representation learning. *NAR Genomics and Bioinformatics*, 4(1):lqac012, January 2022. ISSN 2631-9268. doi: 10.1093/nargab/lqac012.

[27] Naima Vahab and Sonika Tyagi. Advancing non-coding RNA annotation with RNA sequence foundation models: Structure and function perspectives. *BMC Artificial Intelligence*, 1(1): 12, October 2025. ISSN 3005-1924. doi: 10.1186/s44398-025-00012-7.

[28] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch et al. A Kernel Two-Sample Test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.

[29] T.F. Smith and M.S. Waterman. Identification of common molecular subsequences. *Journal of Molecular Biology*, 147(1):195–197, March 1981. ISSN 00222836. doi: 10.1016/0022-2836(81)90087-5.

[30] Ronny Lorenz, Stephan H. Bernhart, Christian Höner zu Siederdissen et al. ViennaRNA Package 2.0. *Algorithms for Molecular Biology*, 6(1):26, 2011. doi: 10.1186/1748-7188-6-26.

[31] Eric W Sayers, Evan E Bolton, J Rodney Brister et al. Database resources of the national center for biotechnology information. *Nucleic Acids Research*, 50(D1):D20–D26, January 2022. ISSN 0305-1048, 1362-4962. doi: 10.1093/nar/gkab1112.

[32] W. James Kent, Charles W. Sugnet, Terrence S. Furey et al. The Human Genome Browser at UCSC. *Genome Research*, 12(6):996–1006, June 2002. ISSN 1088-9051, 1549-5469. doi: 10.1101/gr.229102.

[33] Olga Dudchenko, Sanjit S. Batra, Arina D. Omer et al. De novo assembly of the Aedes aegypti genome using Hi-C yields chromosome-length scaffolds. *Science*, 356(6333):92–95, 2017. doi: 10.1126/science.aal3327.

[34] Jennifer Harrow, Adam Frankish, Jose M. Gonzalez et al. GENCODE: The reference human genome annotation for The ENCODE Project. *Genome Research*, 22(9):1760–1774, September 2012. ISSN 1088-9051. doi: 10.1101/gr.135350.111.

[35] Patricia P Chan, Brian Y Lin, Allysia J Mak et al. tRNAscan-SE 2.0: Improved detection and functional classification of transfer RNA genes. *Nucleic Acids Research*, 49(16):9077–9096, September 2021. ISSN 0305-1048, 1362-4962. doi: 10.1093/nar/gkab688.

[36] Jeffrey J. Quinn and Howard Y. Chang. Unique features of long non-coding RNA biogenesis and function. *Nature Reviews Genetics*, 17(1):47–62, January 2016. ISSN 1471-0056, 1471-0064. doi: 10.1038/nrg.2015.10.

[37] Jeremy E. Wilusz, Courtney K. JnBaptiste, Laura Y. Lu et al. A triple helix stabilizes the 3′ ends of long noncoding RNAs that lack poly(A) tails. *Genes & Development*, 26(21): 2392–2407, November 2012. ISSN 0890-9369, 1549-5477. doi: 10.1101/gad.204438.112.

[38] Tomohiro Yamazaki, Sylvie Souquere, Takeshi Chujo et al. Functional Domains of NEAT1 Architectural lncRNA Induce Paraspeckle Assembly through Phase Separation. *Molecular Cell*, 70(6):1038–1053.e7, June 2018. ISSN 10972765. doi: 10.1016/j.molcel.2018.05.019.

[39] Jesse M. Engreitz, Amy Pandya-Jones, Patrick McDonel et al. The Xist lncRNA Exploits Three-Dimensional Genome Architecture to Spread Across the X Chromosome. *Science*, 341(6147):1237973, August 2013. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.1237973.

[40] Jenny Chen, Alexander A. Shishkin, Xiaopeng Zhu et al. Evolutionary analysis across mammals reveals distinct classes of long non-coding RNAs. *Genome Biology*, 17(1):19, December 2016. ISSN 1474-760X. doi: 10.1186/s13059-016-0880-9.

[41] Wenze Huang, Tuanlin Xiong, Yuting Zhao et al. Computational prediction and experimental validation identify functionally conserved lncRNAs from zebrafish to human. *Nature Genetics*, 56(1):124–135, January 2024. ISSN 1061-4036, 1546-1718. doi: 10.1038/s41588-023-01620-7.

[42] Tao Shen, Zhihang Hu, Siqi Sun et al. Accurate RNA 3D structure prediction using a language model-based deep learning approach. *Nature Methods*, 21(12):2287–2298, December 2024. ISSN 1548-7091, 1548-7105. doi: 10.1038/s41592-024-02487-0.

[43] Bogdan Kirilenko. Curia cross-species ncrna correspondence predictions across 19 mammalian genomes, 2026. URL `https://doi.org/10.5281/zenodo.21383175`.

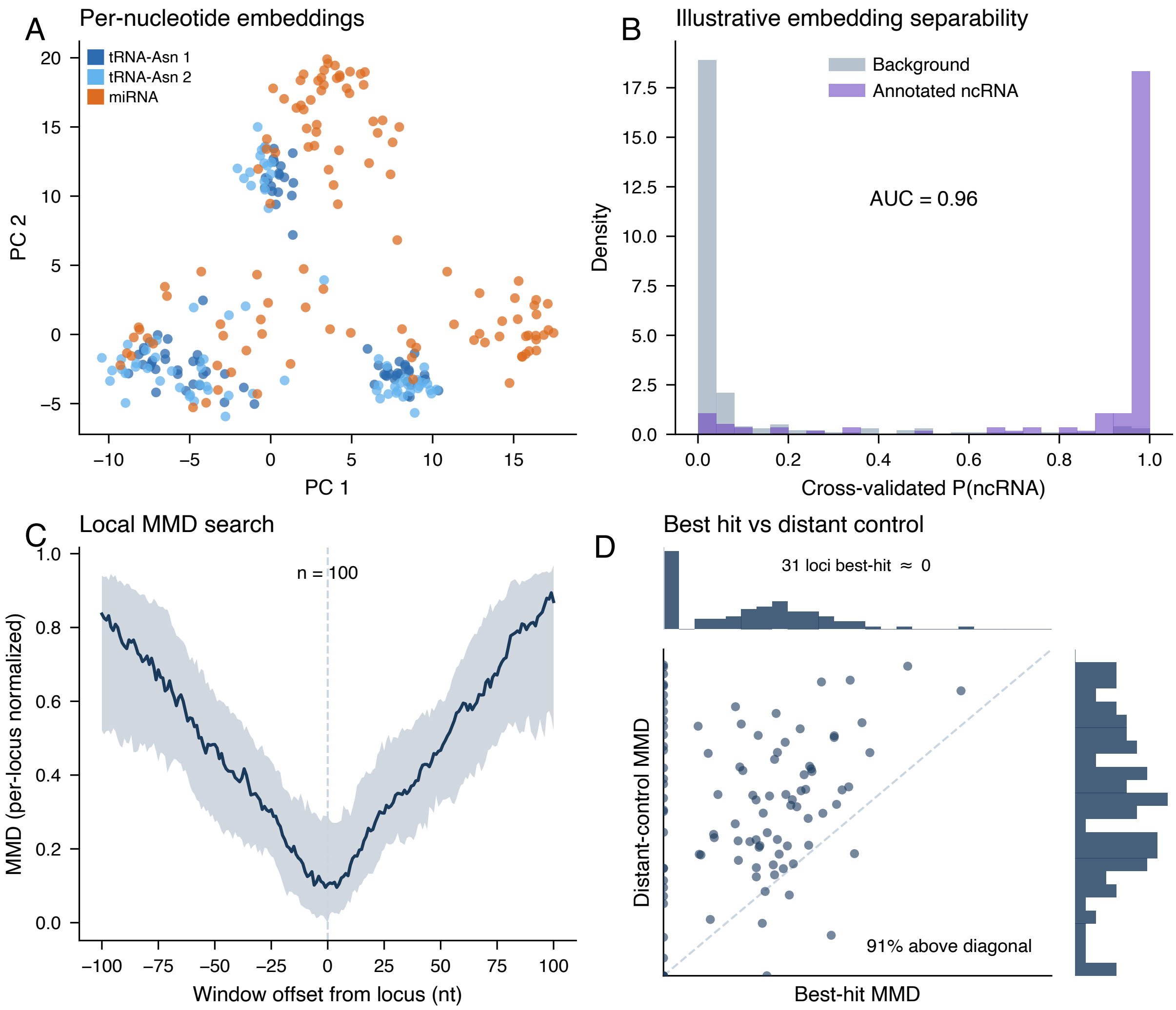

Figure 1: Illustrative embedding behavior supporting local ncRNA comparison. **(A)** Per-nucleotide embeddings (matching PCA, first two components) for two human tRNA-Asn-GTT gene copies (*tRNA-Asn-GTT-chr1-139* and *-140*, 74 nt each) and one pre-miRNA (*MIR103A1*, ENST00000362168, 110 nt), all from the human genome assembly hg38; each point is a nucleotide. The tRNA copies cluster together while the miRNA forms a distinct cluster. **(B)** Out-of-fold probability from a 5-fold cross-validated logistic classifier trained on the embeddings to distinguish annotated hg38 ncRNAs (purple; tRNA, snoRNA, miRNA) from background sequences (gray; intergenic and dinucleotide-shuffled): ncRNAs score near 1 and background near 0 (AUC ≈0.96), showing the embeddings separate the selected RNA-derived sequences from genomic background. Panel (B), titled "Illustrative embedding separability" within the figure, is an illustrative separability experiment on a small balanced set and is not the production island detector used in the pipeline (Section 2.7), which is trained on a different biotype set and negatives and operates at a different threshold. **(C)** Maximum Mean Discrepancy (MMD) between a short-ncRNA reference and windows tiled across its extended locus, versus offset from the annotated position (median and interquartile range over $n = 100$ short ncRNAs, each locus min–max normalized). MMD is minimised at the true position (offset 0) and rises with positional shift. **(D)** Best-hit MMD versus a distant control window (250 nt away) for the same $n = 100$ loci, with marginal histograms. Best-hit MMD concentrates near 0 (31 loci essentially perfect), whereas the distant control almost never does; 91% of loci lie above the diagonal (matched window more similar than the control).

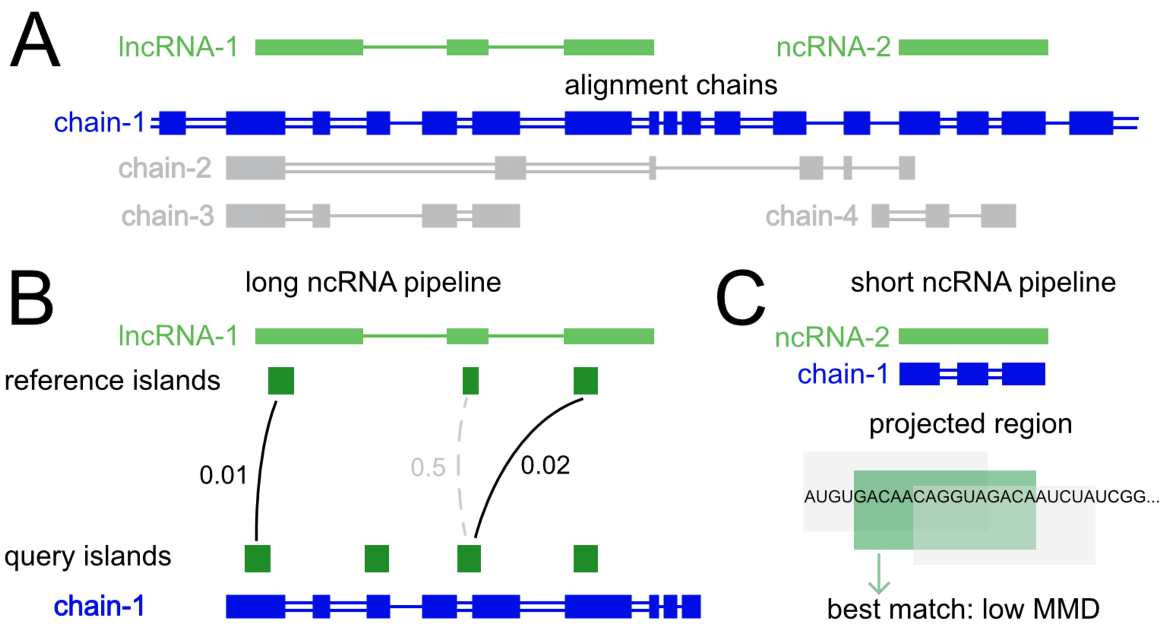

Figure 2: Overview of the CURIA pipeline for cross-species ncRNA correspondence analysis. **(A)** Genome alignment chains restrict the search space by defining candidate loci between reference and query genomes. **(B)** For island-branch loci, localized subregions ("islands") are detected and matched across species using embedding-based similarity, enabling localization of candidate corresponding regions within transcripts. **(C)** For short ncRNAs, correspondence is resolved via local boundary refinement using sliding-window matching in embedding space.

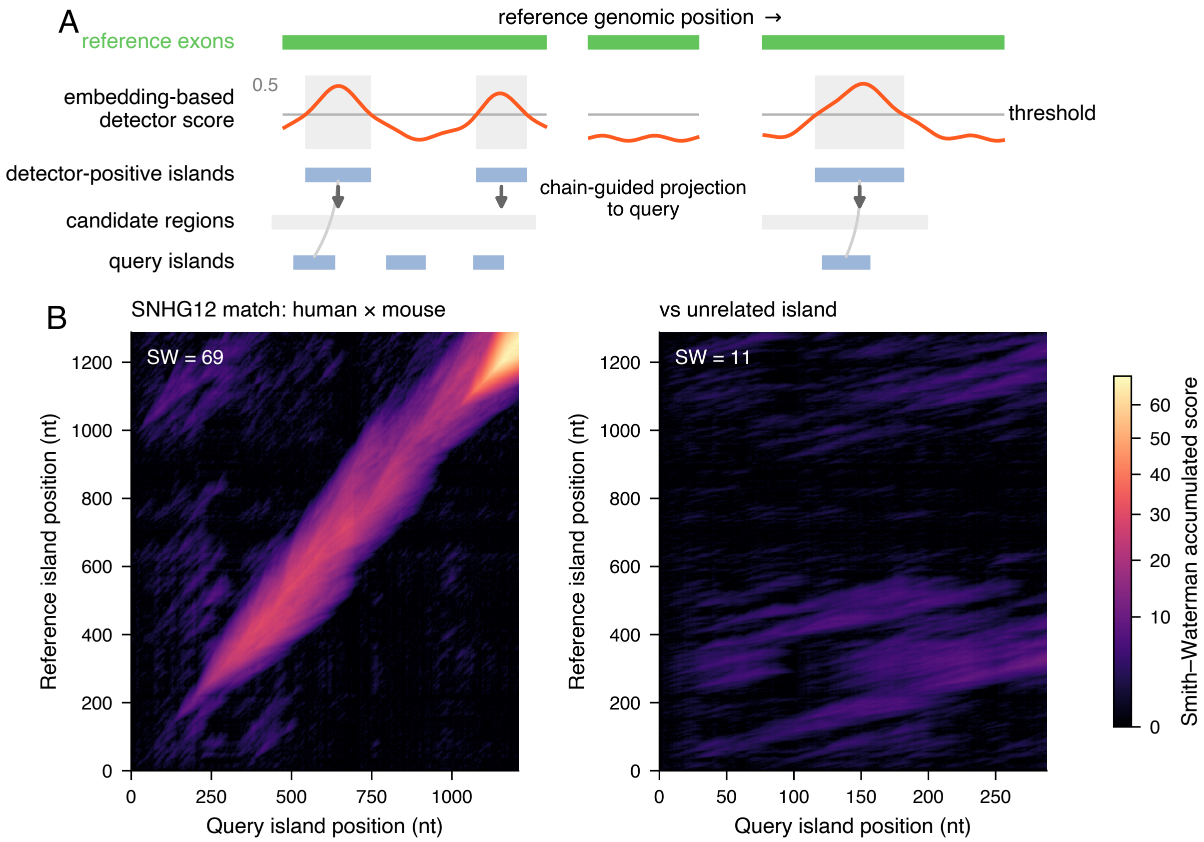

Figure 3: Identification and matching of localized ncRNA subregions using embedding-based similarity. **(A)** Schematic: an embedding-based detector score along a reference transcript is thresholded to define localized subregions (“islands”); islands are projected via genome alignment chains to candidate query regions, where corresponding islands are detected independently. **(B)** Island matching for a recurrent lncRNA region (SNHG12, human hg38 chr1 × mouse mm39 chr4). Each island is embedded once; the per-token cosine dotplot is aligned by Smith–Waterman, shown here as the accumulated-score matrix. The conserved matched region forms a single coherent diagonal ridge (embedding-SW alignment score 69), whereas the same human island against an unrelated mouse island yields no coherent alignment (embedding-SW alignment score 11).

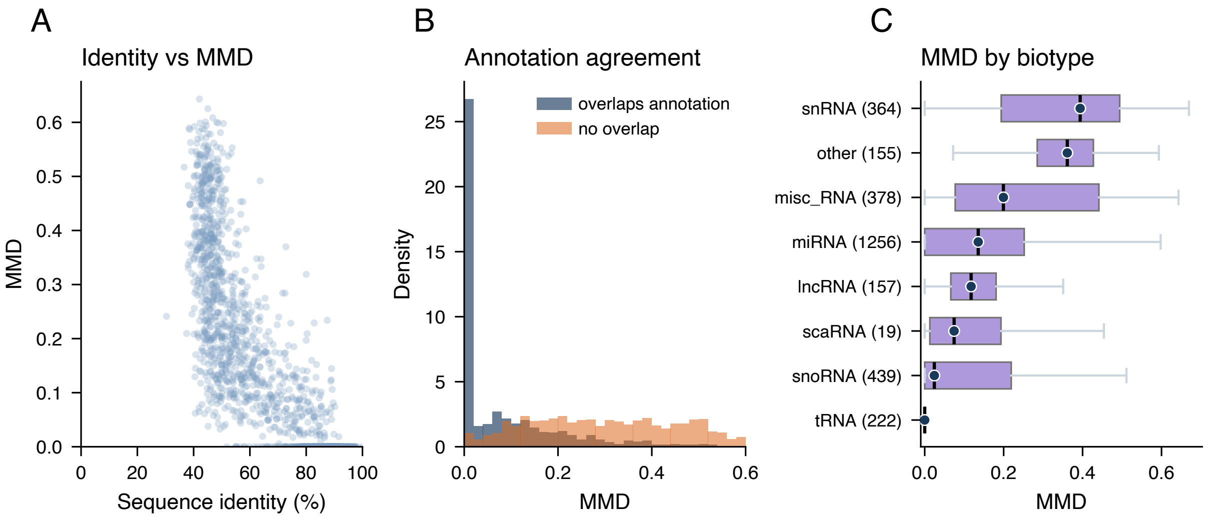

Figure 4: Behavior of embedding-based similarity (MMD) across short ncRNA loci for the human–mouse pair. **(A)** Sequence identity versus MMD for refined predicted loci. **(B)** MMD distributions for loci that do versus do not overlap mouse annotation: annotation-overlapping loci concentrate at low MMD, non-overlapping loci shift higher. **(C)** MMD distribution by ncRNA biotype, ordered by median.

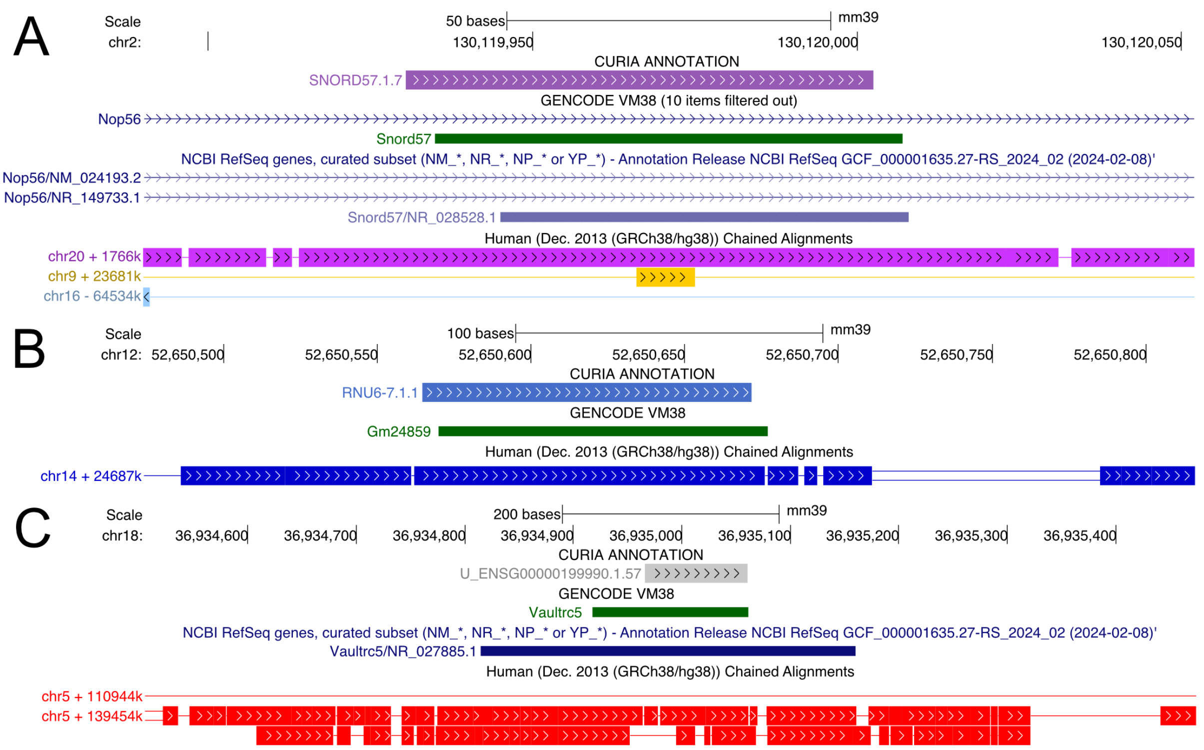

Figure 5: Representative compact ncRNA projections from human (`hg38`) to mouse (`mm39`). Genome-browser views show a localized SNORD57 match overlapping *Snord57*, an RNU6-7 prediction intersecting *Gm24859*, and a vault-RNA prediction overlapping *Vaultrc5* with different interval boundaries. SNORD57 and RNU6-7 had edit-derived identities of 74% and 93%, respectively, with reported MMD values of 0.00. The vault-RNA match had 59% edit-derived identity, normalized nucleotide-SW score 0.35, and MMD 0.14.

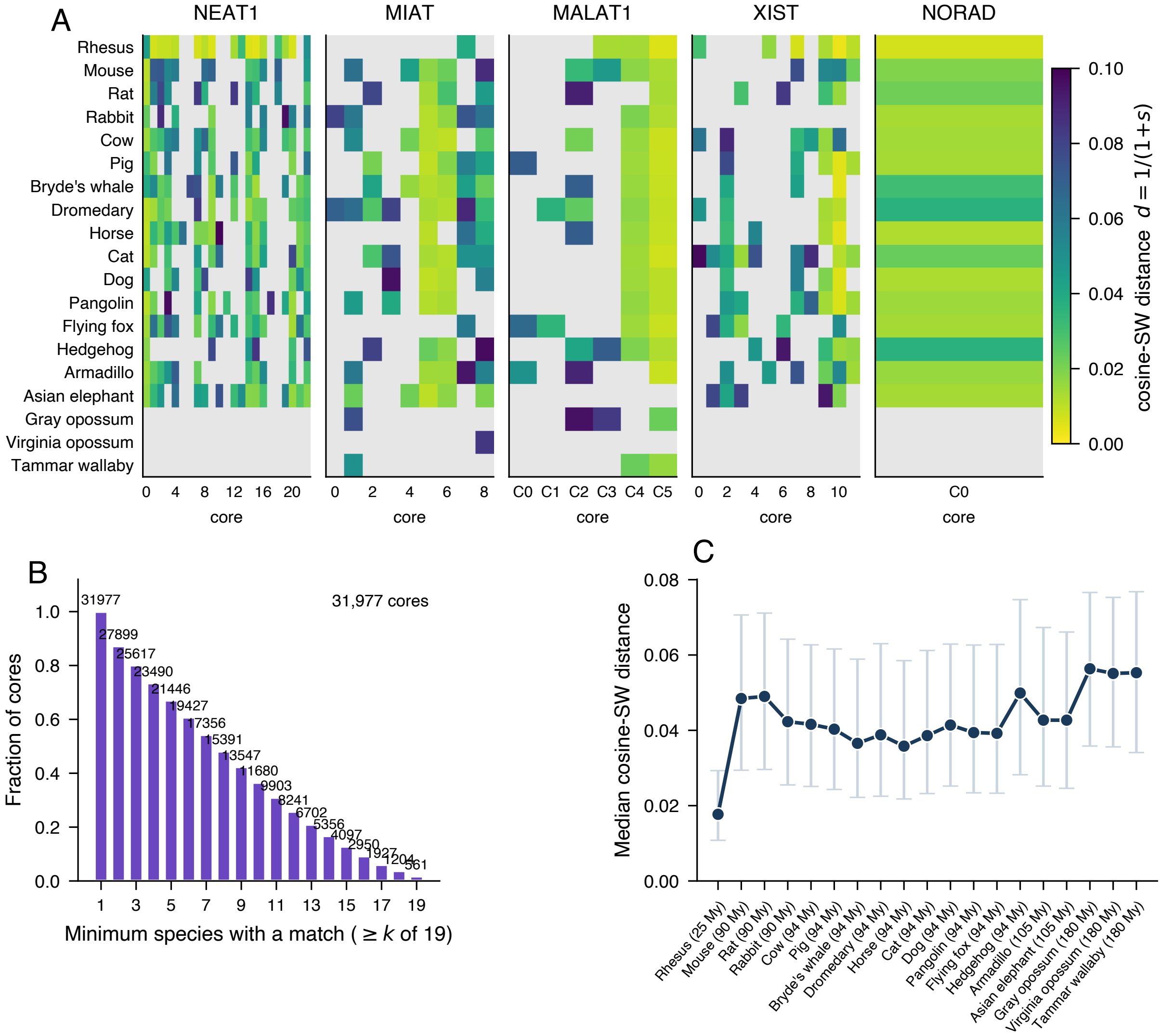

Figure 6: Cross-species recurrence and match strength of localized embedding-supported regions. **(A)** Accepted matches for reference cores in representative lncRNA loci (NEAT1, MIAT, MALAT1, XIST, and NORAD). Rows represent query species, columns represent reference cores ordered along the human locus, and color indicates embedding-SW distance $d = 1/(1+s)$. **(B)** Distribution of the number of query genomes containing an accepted match for each aggregated reference-coordinate core. **(C)** Median embedding-SW distance across accepted matches for each query genome. This descriptive score is length-dependent and does not show a strictly monotonic relationship with phylogenetic distance. Panel **(B)** is conditional on the 31,977 aggregated reference-coordinate cores with at least one accepted match. Loci without a detected reference island and islands without an accepted match are excluded from this denominator. The set contains 30,926 cores from loci annotated as lncRNA and 1,051 from other biotypes processed by the localized-subregion workflow. Both unsuccessful projection and failure to obtain an accepted match appear as an absent species.